

a. Joan 有女儿

$$\exists x (\text{Parent}(\text{Joan}, x) \wedge \text{Female}(x))$$

b. Joan 只有一个女儿

$$\exists^1 x (\text{Parent}(\text{Joan}, x) \wedge \text{Female}(x))$$

c. Joan 只有一个孩子，是女儿

$$\exists^1 x (\text{Parent}(\text{Joan}, x)) \wedge \text{Female}(x)$$

d. Joan 和 Kevin 只有一个儿子

$$\exists^1 x (\text{Parent}(\text{Joan}, x) \wedge \text{Parent}(\text{Kevin}, x) \wedge \neg \text{Female}(x))$$